Modular Quotient Observations: Correlated Remainders, Randomized Lifting, and Rényi Security Transfers

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Abstract

Modular quotient constructions combine exact arithmetic identities with distributional questions that must be analyzed separately. This paper develops a general distributional framework for modular quotient observations, including structured vector observations. It distinguishes an exact additive representation of a discarded remainder from a law of independent noise. Uniform randomized lifting is characterized by the conditional distribution inside each quotient cell, with all public side information retained. A quotient-only universal simulator of a fixed additive-noise channel is shown to require uniform noise modulo the ambient modulus; approximate simulation with broader noise is nevertheless possible under an explicit loss. Subtractive dither illustrates why independence after discarding a variable need not survive its publication. For overlapping Hankel windows, a sequential conditioning argument bounds the exponentiated order-two Rényi divergence between deterministic rounded observations and independently perturbed rounded observations. Exact cell costs refine this bound. The resulting recovery-success transfer is distinguished from an additive decision guarantee and from a complete encryption-security proof. All arguments retain their declared distributions and are accompanied by exact finite verification. The result is a distributional toolkit for modular quotient constructions: it does not establish a reduction from standard LWE, a complete encryption-security proof, or concrete post-quantum parameters.

Scope. The paper is self-contained at the level of its numbered statements. Classical Rényi-divergence and dither antecedents are cited publicly; the security theorem for middle-product LWR is used only for attribution of a probabilistic-lifting convention and is not transferred to the present model. No claim of IND-CPA, IND-CCA, 128-bit security, or a hardness reduction follows from the distributional identities below.

1. A finite model with an explicit public transcript

Let P, Q be positive integers. Unless a narrower condition is stated, the cell and lifting identities allow any such pair, including $Q > P$. Define

$$[z]_P = z - P\lfloor z/P \rfloor, \qquad F(z) = F_{P,Q}(z) = \left\lfloor \frac{[z]_P}{Q} \right\rfloor, \qquad J = \lceil P/Q \rceil. \tag{1}$$

Residues use canonical integer representatives. For $0 \le u < J$, put

$$\omega_u = \min\{Q, P - Qu\}, \qquad I_u = \{Qu, \dots, Qu + \omega_u - 1\}. \tag{2}$$

All cells are nonempty. The last one may be shorter than Q. Quotient and remainder maps apply componentwise to vectors.

Let A denote public information and X a secret, both with finite support. A deterministic, integer-valued map f gives

$$Z = [f(A, X)]_P \in \mathbb{Z}_P^L, \qquad U = F(Z), \qquad \rho = Z - QU, \tag{3}$$

where $L \geq 1$. This is a publication model, not yet an encryption algorithm. An auxiliary observation may be $[f(A, X) + E]_P$, with the joint law of E, A, X specified separately.

All statistical distances below are distances between finite laws. Integer noise with countable support is allowed when its residue distribution is well defined; the structured bounds additionally state positivity or moment hypotheses. A variable included for analysis need not be supplied to an adversary. In particular, comparing (A, X, U) is useful for a true-secret recovery event, while a public distinguisher sees (A, U), not X.

For a randomized test D, the distinguishing advantage is the absolute acceptance gap. In an equiprobable challenge-bit experiment the corresponding optimized success bias is half that gap. Any computational use of a probability kernel additionally requires an efficient implementation. Enumerating a finite distribution does not meet that requirement merely because the distribution exists.

2. The exact arithmetic error, including multiple wraparounds

The deterministic representation

$$QU \equiv f(A, X) - \rho \pmod{P}, \qquad 0 \leq \rho_i < \omega_{U_i} \tag{4}$$

is always correct. It does not assert that $-\rho$ is independent of A, X.

Proposition 1 — General-modulus perturbation identity. For a canonical scalar residue $z = Qu + r$, write

$$e = Qh + \delta, \quad P = aQ + b, \quad 0 \leq r < Q, \quad 0 \leq \delta, b < Q, \qquad k = \lfloor (z + e)/P \rfloor.$$

Then

$$F(z + e) = u + h - ka + \left\lfloor \frac{r + \delta - kb}{Q} \right\rfloor. \tag{5}$$

If $Q \mid P$, the expression simplifies to

$$F(z + e) = \left[u + h + \left\lfloor \frac{r + \delta}{Q} \right\rfloor \right]_{P/Q}. \tag{6}$$

Proof. The canonical residue of $z + e$ is $z + e - kP$. Substitute the displayed Euclidean divisions and take its quotient by Q. No restriction to one wraparound is used. If $b = 0$, reduction modulo $P = Qa$ reduces the quotient modulo a, proving (6). Negative errors are covered by floor division, not truncation toward zero. □

When $Q \nmid P$, the term $-kb$ shifts the final cell boundaries. Omitting it can change the quotient. The proposition is an identity for given integers; it assigns no probability to a carry and gives no independence claim.

3. Marginal remainder laws and conditioning on the public

Proposition 2 — Uniform-input cell laws. Suppose Z is uniform on $\mathbb{Z}_P$, and write $P = aQ + b$, $0 \le b < Q$. For $U = F(Z)$ and $R = Z - QU$,

$$\Pr[U = u] = \frac{\omega_u}{P}, \qquad \Pr[R = r \mid U = u] = \frac{\mathbf{1}_{0 \le r < \omega_u}}{\omega_u}, \tag{7}$$

$$\Pr[R = r] = \frac{a + \mathbf{1}_{r<b}}{P} \quad (0 \le r < Q), \qquad \mathrm{TV}(\mathcal{L}(R), \mathrm{Unif}(\mathbb{Z}_Q)) = \frac{b(Q-b)}{PQ}. \tag{8}$$

If $b = 0$, U, R are independent uniforms. If $b > 0$,

$$\mathrm{TV}(\mathcal{L}(U), \mathrm{Unif}(\{0, \dots, J-1\})) = \frac{a(Q-b)}{PJ}. \tag{9}$$

Proof. Every canonical residue has a unique quotient and remainder. A cell has ω_u equiprobable points, proving (7). Each remainder occurs in a complete blocks and once more exactly when $r < b$. The positive deviations from $1/Q$ occur at those b positions; their sum is (8). For $b > 0$, $J = a + 1$, each complete cell has mass Q/P, and the short cell has mass b/P. Sum the positive deviations from $1/J$. If $a = 0$, there are no complete cells, and the quotient is constant, so (9) is zero. □

Uniformity of Z is an assumption, not a consequence of writing a matrix product. Even under that assumption, independence from the quotient is not independence from all the public information. In (3), ρ is deterministic given (A, X).

For a uniform finite secret set $\mathcal{X}$ independent of A, the conditional remainder law at a positive-probability publication is

$$\Pr[\rho = r \mid A = a, U = u] = \frac{\#\{x \in \mathcal{X} : F(f(a,x)) = u,\ [f(a,x)]_P = Qu + r\}}{\#\{x \in \mathcal{X} : F(f(a,x)) = u\}}. \tag{10}$$

This is a counting identity, not an efficient multidimensional counting algorithm. Nonuniform secret priors require their corresponding weights in numerator and denominator.

A correlated-remainder witness. For the Hankel map defined by the displayed matrix, take $P = 257, Q = 15, L = 3, d = 2$, ribbon

$$a = (257, 258, 257, 259), \qquad \mathsf{H}_{3,2}(a) \bmod P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 2 \end{pmatrix},$$

and independent uniform $X_0, X_1 \in \{8, \dots, 15\}$. Conditional on $U = (0, 0, 1)$, the compatible secrets are exactly $\{8, \dots, 14\}^2$, and

$$\rho = (X_1, X_0, 2X_1 - 15). \tag{11}$$

There are 49 equally likely joint values, while the product of the three marginal laws has 343. Each marginal is uniform on seven points, but the first and third coordinates determine one another. Directly summing the masses gives distance $6/7$ from that product law. This explicit witness is included only to show that coordinatewise uniform marginals need not imply a product law; it is not evidence that such a public ribbon is typical or a claimed attack on every structured construction.

4. Randomized lifting preserves cells, not arbitrary hidden residues

Given u, the uniform lifting kernel samples a residue uniformly from I_u. For a vector it samples the product cell $I_{u_0} \times \cdots \times I_{u_{L-1}}$. It requires no knowledge of the original secret.

Theorem 3 — Exact criterion for a cell-uniform lift. Let (C, Z) have any finite joint law, let $U = F(Z)$, and, conditional on (C, U), draw $\widetilde{Z}$ uniformly from the product cell I_U, independently of the old Z. Then

$$F(\widetilde{Z}) = U,$$

$$\mathrm{TV}(\mathcal{L}(C, Z), \mathcal{L}(C, \widetilde{Z})) = \sum_{c,u} \Pr[C = c, U = u]\ \mathrm{TV}\big(\mathcal{L}(Z \mid c, u), \mathrm{Unif}(I_u)\big). \tag{12}$$

Only pairs (c, u) of positive probability are included. Consequently the lift reconstructs the joint law exactly if and only if all these conditional laws are uniform on their entire product cells. Lifting a second time has no further effect on its output law.

Proof. Both experiments have the same (C, U) marginal, because their residues stay in the same cells. Distinct pairs (c, u) have disjoint supports in (c, z). Split the total-variation sum over these supports, factor out their common masses, and obtain (12). Every term is nonnegative, so equality to zero is equivalent to conditional uniformity on each positive-mass cell. The lifted law already has that property, proving idempotence. □

For $Q \mid P$, write $H = P/Q$. If U is uniform on $\mathbb{Z}_H^L$ independently of C, lifting produces a uniform residue vector independently of C. More generally, if C's law is unchanged, the disjoint-cell calculation gives

$$\mathrm{TV}\big((C, \widetilde{Z}), \mathcal{L}(C) \otimes \mathrm{Unif}(\mathbb{Z}_P^L)\big) = \mathrm{TV}\big((C, U), \mathcal{L}(C) \otimes \mathrm{Unif}(\mathbb{Z}_H^L)\big). \tag{13}$$

This can justify a lift in an ideal game. It does not justify declaring the actual lifted product uniform in a real game. When $Q \nmid P$, the correct quotient law of a uniform residue vector is the product of the size-weighted cell laws in (7), not a uniform quotient alphabet.

For a fixed scalar input $z = Qu + r$, the integer error of a uniform lift is $\widetilde{Z} - z = R' - r$, where R' is uniform on $\{0, \ldots, \omega_u - 1\}$. Its support depends on r. Even when Z is uniform and $Q \mid P$, the triangular marginal law

$$\Pr[R' - R = e] = \frac{Q - |e|}{Q^2} \quad (|e| < Q) \tag{14}$$

does not make this integer error independent of Z when $Q > 1$: its conditional support shifts with the true remainder. For $Q = 1$, the error is identically zero and is independent in the degenerate sense. This formula concerns the integer difference of remainders, not an unrestricted reinterpretation of its reduction modulo P.

5. A precisely scoped obstruction to universal dequantization

Theorem 4 — Quotient-only simulation of an additive channel. Let $2 \le Q \le P$. Suppose a single randomized kernel K, receiving only $F(z)$, must satisfy

$$K(F(z)) \overset{d}{=} [z + E]_P \quad \text{for every } z \in \mathbb{Z}_P, \tag{15}$$

where the noise law modulo P is the same for every input. Then that noise law is uniform modulo P. Conversely, a kernel returning a fresh uniform residue satisfies (15).

For any proposed approximate kernel and two residues z, z' in one cell,

$$\max\{\mathrm{TV}(K(F(z)), \chi_z), \mathrm{TV}(K(F(z')), \chi_{z'})\} \geq \tfrac{1}{2}\,\mathrm{TV}(\chi_z, \chi_{z'}), \tag{16}$$

where $\chi_z = \mathcal{L}([z + E]_P)$.

Proof. The residues 0 and 1 have the same quotient. Their output laws in (15) must therefore be equal. The noise law is invariant under translation by one, hence under every translation of the finite cyclic group, and must be uniform. Uniform output proves the converse. For (16), the two kernel laws are one and the same law. Apply the triangle inequality through it and bound the sum by twice its larger summand. □

The quantifiers are essential. This theorem does not exclude a transport that also uses A, other observations, or an input distribution; it does not exclude a change of dimensions or a reduction that transfers advantage rather than exact samples. It also does not require recovering the original input. For $Q = 1$, the quotient retains the entire residue and the obstruction does not apply. Uniform noise is a degenerate solution, not a useful confidentiality theorem.

6. Approximate simulation by a wider error distribution

Define the integer center of a cell by

$$c(u) = Qu + \lfloor (\omega_u - 1)/2 \rfloor. \tag{17}$$

For a vector of quotient observations, independently sample E_i uniformly on $\{-T, \ldots, T\}$, $T \geq 0$, and output $\widetilde{Y}_i = [c(U_i) + E_i]_P$.

Proposition 5 — Conditional smoothing bound. For any joint law of (C, Z), with $U = F(Z)$, and product noise independent of (C, Z),

$$\mathrm{TV}\big((C, Z, \widetilde{Y}), (C, Z, [Z + E]_P)\big) \leq \mathbb{E} \min\left\{1, \sum_{i=0}^{L-1} \frac{\lfloor \omega_{U_i}/2 \rfloor}{2T+1}\right\} \leq \min\left\{1, \frac{L\lfloor Q/2 \rfloor}{2T+1}\right\}. \tag{18}$$

The same bound holds after hiding Z.

Proof. The maximum distance from an integer point of a cell to its chosen center is $\lfloor \omega_u/2 \rfloor$. Two integer uniform laws on intervals of common size $2T + 1$, shifted by s, have total variation $\min\{1, |s|/(2T + 1)\}$, by counting the overlap of their supports. Modular reduction cannot increase this distance. Conditional on (C, Z), couple each coordinate optimally; a union bound bounds the probability that any coordinate differs by the sum of the coordinate distances, capped at one. Average over the common conditioning variables. Finally, forgetting variables is a common deterministic map and contracts total variation. □

Corollary 6 — An exact scalar minimax value for an odd complete cell. Suppose $Q = 2h + 1$, $T \geq h$, and $P \geq 2T + Q$. For a complete cell of Q points and noise uniform on $[-T, T] \cap \mathbb{Z}$, the smallest possible worst-input error of a quotient-only simulator is

$$\inf_K \sup_{z \text{ in the cell}} \mathrm{TV}(K(F(z)), [z + E]_P) = \frac{h}{2T+1}. \tag{19}$$

Proof. If $h = 0$, the cell is a singleton and exact simulation is possible, giving value zero. Assume now $h \geq 1$. Noise laws centered at the cell endpoints differ by $2h$. The union of their integer support intervals has at most $2T + Q \leq P$ points, so reduction modulo P creates no additional overlaps. Their distance is $2h/(2T + 1)$. Theorem 4 gives half this value as a necessary worst-case

error. The integer center realizes at most $h/(2T+1)$ for every cell point by Proposition 5. □

This simulator receives the quotient, not the secret or true remainder. Uniform integer sampling has an exact expected-polynomial implementation in the bit length of T; a strict cap needs an explicit failure allowance. A large T can make the modular noise nearly uniform and can destroy useful information. The direction is also important: a solver for the broader-noise experiment, composed with this simulator, would solve a rounded instance approximately. This is not automatically the desired hardness reduction from an already founded source to MODDIV.

7. Subtractive dither and an observable residual relation

The classical dither literature motivates this calculation [R3], but the following finite-modular statement is proved directly and its full transcript is kept explicit.

Theorem 7 — Independence after forgetting dither is not full-transcript independence. Suppose $Q \mid P$. For any random residue Z, independently sample T uniformly on $\{0, \dots, Q-1\}$, and set

$$Y = [QF(Z+T) - T]_P. \tag{20}$$

Then $Y = [Z - R]_P$, where $R = [Z+T]_Q$ is uniform on $\mathbb{Z}_Q$ independently of Z. However, let R' instead be independent of (Z,T), with that same uniform law, and put $Y' = [Z - R']_P$. Then

$$\mathrm{TV}((Z,T,Y),(Z,T,Y')) = \mathrm{TV}((T,Y),(T,Y')) = 1 - \frac{1}{Q}. \tag{21}$$

Proof. Since $Q \mid P$, the remainder of $[Z+T]_P$ modulo Q equals $[Z+T]_Q$. Quotient–remainder reconstruction gives (20) as $Z - R$ modulo P. At each fixed $Z = z$, translation by z permutes the Q possible values of T, proving uniformity and independence of R from Z.

For each fixed (z,t), the real Y is deterministic. The comparison law Y' is uniform on Q distinct residues and includes that deterministic value, so its conditional distance is $1 - 1/Q$. Averaging proves the first equality in (21). The public event $[Y+T]_Q = 0$ holds with probability one in the real experiment and $1/Q$ in the comparison, because R' is independent uniform. It supplies the same lower bound after hiding Z; data processing gives the upper bound. The statement also includes $Q = 1$, where the distance is zero. □

This does not forbid using dither in an explicitly specified protocol. It forbids replacing its full public transcript by an independent-noise transcript without proving the replacement. Moreover, $F(Z+T)$ cannot generally be computed from $F(Z)$ and T alone. No dither is inserted into any existing MODDIV construction by this article.

8. The same efficient transformation must work in both worlds

For finite laws μ, ν, use

$$\mathrm{TV}(\mu,\nu) = \tfrac{1}{2}\sum_z |\mu(z) - \nu(z)|.$$

A probability kernel $K(y \mid x)$ is nonnegative and each input row sums to one; μK denotes its output law.

Lemma 8 — Kernel contraction and two-world transfer. A common kernel satisfies

$$\mathrm{TV}(\mu K, \nu K) \le \mathrm{TV}(\mu,\nu). \tag{22}$$

Suppose source laws P_b, target laws Q_b, and the same efficient kernel satisfy

$$\mathrm{TV}(\mathsf{P}_b K, \mathsf{Q}_b) \le \varepsilon_b \quad (b = 0, 1).$$

Then every efficient target distinguisher D induces $B = D \circ K$ with

$$\mathrm{Adv}_{\mathsf{Q}}(D) \le \mathrm{Adv}_{\mathsf{P}}(B) + \varepsilon_0 + \varepsilon_1. \tag{23}$$

Proof. Expand the output masses and apply the triangle inequality to their difference. Summing each kernel row yields (22). For (23), the acceptance probabilities of D, including its independent coins, differ by at most ε_b in each world. Apply the triangle inequality to their two gaps. The composed procedure cannot ask which world generated its input. □

All public values are part of these laws. An argument comparing only hidden products or one-coordinate marginals cannot substitute for (23). If a noisy linear source is already justified and its public and secret distributions are exactly the required ones, applying the same rounding map gives a valid reduction to its rounded image. The rounded null distribution must also be the correct image: uniform quotients when $Q \mid P$, size-weighted cells otherwise. This is not a proof that the required noisy source is difficult.

9. Rényi comparison: preservation of success rather than equality of laws

Define the **exponentiated** order-two divergence

$$\mathcal{R}_2(\mu\|\nu) = \sum_{z:\mu(z)>0} \frac{\mu(z)^2}{\nu(z)}, \tag{24}$$

with value $+\infty$ if a denominator vanishes. It equals $\exp(D_2(\mu\|\nu))$ under the natural-log convention of [R1]; it is the convention used in [R2, Definition 2]. It is not itself a logarithm or a metric.

Lemma 9 — Probability preservation and data processing. If $\mu \ll \nu$, then for every event S,

$$\nu(S) \ge \frac{\mu(S)^2}{\mathcal{R}_2(\mu\|\nu)}. \tag{25}$$

A common kernel satisfies

$$\mathcal{R}_2(\mu K\|\nu K) \le \mathcal{R}_2(\mu\|\nu). \tag{26}$$

If the two laws have the same marginal π on C, then

$$\mathcal{R}_2(\mu_{C,Z}\|\nu_{C,Z}) = \sum_{c:\pi(c)>0} \pi(c)\, \mathcal{R}_2(\mu_{Z|c}\|\nu_{Z|c}). \tag{27}$$

Proof. Cauchy–Schwarz applied to $\sum_{z\in S}(\mu(z)/\sqrt{\nu(z)})\sqrt{\nu(z)}$ gives (25). For each output y, apply the same inequality with weights $\nu(x)K(y \mid x)$:

$$\frac{(\sum_x \mu(x)K(y \mid x))^2}{\sum_x \nu(x)K(y \mid x)} \le \sum_x \frac{\mu(x)^2 K(y \mid x)}{\nu(x)}.$$

Zero-mass terms are omitted. Sum over y to obtain (26). Equation (27) follows by substituting $\mu(c, z) = \pi(c)\mu(z \mid c)$ and the analogous expression for ν. □

Appending the same independent algorithm coins to both laws leaves $\mathcal{R}_2$ unchanged; the lemma therefore applies to a randomized algorithm's success event. There is also the elementary estimate

$$\mathrm{TV}(\mu,\nu) \le \tfrac{1}{2}\sqrt{\mathcal{R}_2(\mu\|\nu)-1}, \tag{28}$$

obtained by applying Cauchy–Schwarz to $\sum_z \nu(z)|\mu(z)/\nu(z)-1|$. A constant divergence does not imply negligible distance. For example, $\mu=(3/4,1/4)$ and $\nu=(1/2,1/2)$ have $\mathcal{R}_2=5/4$ and total variation $1/4$.

The proofs above are elementary finite-space proofs. They do not depend on the general-measure arguments corrected in the 2024 errata to [R1]. No source's notation or rounding convention is silently substituted for (1).

10. The exact joint cost of keeping a rounded publication

For a scalar error law χ, define

$$p_\chi(z)=\Pr_{e\leftarrow\chi}[F(z+e)=F(z)], \qquad V_\chi=\sum_{z=0}^{P-1}\left(\frac{1}{p_\chi(z)}-1\right). \tag{29}$$

The latter is finite if every retention probability is positive; $\Pr[e=0]>0$ is a sufficient condition.

Theorem 10 — Joint divergence for deterministic and noisy rounding. Let (A,X) have the same law in two experiments, with $Z=[f(A,X)]_P$. Write

$$\mathsf{D}_0=(A,X,F(Z)), \qquad \mathsf{D}_\chi=(A,X,F(Z+E)).$$

If, conditional on (A,X), the errors have independent coordinates of law χ, then

$$\mathcal{R}_2(\mathsf{D}_0\|\mathsf{D}_\chi)=\mathbb{E}_{A,X}\prod_{i=0}^{L-1}p_\chi(Z_i)^{-1}. \tag{30}$$

For a general conditional error law the correct expression is instead

$$\mathbb{E}_{A,X}\frac{1}{\Pr[F(Z+E)=F(Z)\mid A,X]}, \tag{31}$$

with infinity if this conditional probability vanishes on an event of positive probability.

Proof. At each fixed (a,x), the noiseless output is a point mass at $F(Z)$. Its divergence from the noisy conditional law is the reciprocal of that law's mass at the same point. Equation (27) gives (31). Conditional independence of the errors factorizes the mass into scalar retention probabilities and gives (30). It does not assert independence of the Z_i. □

Why noise marginals are insufficient. With $P=19, Q=3$ and $Z=(0,0,0)$, take E uniformly from

$$(-1,0,1), \quad (0,1,-1), \quad (1,-1,0). \tag{32}$$

Each coordinate is uniform on $\{-1,0,1\}$, and its individual retention probability is $2/3$. Joint retention is zero because one coordinate always becomes 18 modulo 19, outside the zero cell. Thus (31) is infinite whereas the unjustified product would give $(3/2)^3$.

Support can fail even in one coordinate: with $P=19, Q=3, z=18$ and error uniform on $\{-1,1\}$, the quotient changes from 6 to either 5 or 0. Symmetry of a noise law alone therefore does not guarantee finite divergence. Neither example refutes the theorem under its actual hypotheses.

11. Overlapping Hankel windows and the fresh-coefficient argument

Let $d, L, t \geq 1$, and let each of t independent public ribbons have length $L + d - 1$. Its entries are independent uniforms on an integer interval

$$\mathcal{I} = \{c, \ldots, c + M - 1\}, \qquad 1 \leq M \leq P. \tag{33}$$

The secret vector $X \in \mathbb{Z}^d$ has any finite law independent of all ribbons. The same secret is used for all of them. Define

$$Z_{j,i} = \left[\sum_{k=0}^{d-1} A_{i+k}^{(j)} X_k \right]_P, \quad 1 \leq j \leq t, \quad 0 \leq i < L. \tag{34}$$

Take a public bound

$$\gamma \geq \max_{x \in \mathrm{supp}(X)} \gcd(x_{d-1}, P), \qquad 1 \leq \gamma \leq P. \tag{35}$$

The endpoint is x_{d-1} in this Hankel convention. Reversing the secret gives the equivalent x_0 Toeplitz indexing convention.

Theorem 11 — Structured joint Rényi bound. If all tL errors are independent of each other and of $(\mathbf{A}, X)$, with common law χ and finite V_χ, then

$$\mathcal{R}_2(\mathsf{D}_0 \| \mathsf{D}_\chi) \leq C_\chi := \left(1 + \frac{\gamma V_\chi}{M}\right)^{tL}. \tag{36}$$

Here the distributions include X and every public ribbon, as in Theorem 10.

Proof. Fix $X = x$. Conditional on all coefficients before A_{i+d-1}, a row has the form $x_{d-1}A_{i+d-1} + c_i$ modulo P. A linear congruence in the fresh coefficient has at most $\gcd(x_{d-1}, P)$ solutions in an interval of length at most P, because its canonical solutions are equally spaced by $P/\gcd(x_{d-1}, P)$. Every row residue therefore has conditional mass at most γ/M. This bound is still true, though weak, when that ratio exceeds one.

Set $g(z) = p_\chi(z)^{-1} \geq 1$. Then

$$\mathbb{E}[g(Z_i) \mid x, \text{earlier coefficients}] \leq 1 + \frac{\gamma}{M} \sum_z (g(z) - 1). \tag{37}$$

In the product of L such factors, the last fresh coefficient appears only in the last factor. Integrate it out using (37), and repeat backwards. This gives the L-th power without assuming independent rows. Conditional on x, the public ribbons are independent, so the same procedure applies to all t ribbons. Finally average over X and use (30). □

When $M = P$ and every supported endpoint secret is a unit modulo P, each fresh row residue is exactly uniform conditional on the earlier coefficients. In that special case all the conditional inequalities are equalities and

$$\mathcal{R}_2(\mathsf{D}_0 \| \mathsf{D}_\chi) = \left(1 + \frac{V_\chi}{P}\right)^{tL}. \tag{38}$$

Uniformity here is over the random public ribbons while the secret is fixed. It does not mean that Z remains uniform after revealing those ribbons and the secret. Nor does it certify a difficulty assumption for this structured source.

12. Explicit cell costs for bounded independent noise

Let e be uniform on $\{-B,\dots,B\}$, $B \geq 0$. For a cell of width $a \geq 1$, the number of shifts that stay inside it on the integer line from position j is

$$c_j = 1 + \min(B, j) + \min(B, a-1-j).$$

Set

$$g_B(a) = \sum_{j=0}^{a-1} \left(\frac{2B+1}{c_j} - 1 \right), \qquad V_B^\sharp = (J-1)g_B(Q) + g_B(\omega_{J-1}). \tag{39}$$

Proposition 12 — Cell-retention bound and its elementary envelope. For $1 \leq Q \leq P$,

$$V_B \leq V_B^\sharp. \tag{40}$$

Equality holds if $B \leq P - Q$. Furthermore,

$$g_B(a) = 2B + 1 - a \quad (a \leq B+1), \qquad g_B(a) \leq B \quad (a \geq B+1). \tag{41}$$

If $a \geq 2B+1$, then

$$g_B(a) = \psi_B := 2\left((2B+1) \sum_{j=B+1}^{2B} \frac{1}{j} - B \right), \tag{42}$$

where the sum is empty and $\psi_0 = 0$ for $B = 0$.

Proof. Every shift counted by c_j remains in the same modular cell. Wrapping can only add such shifts, so $p_B(z) \geq c_j/(2B+1)$. Summing proves (40). To leave a cell of length $a \leq Q$, wrap around, and enter that cell again requires an integer shift of magnitude at least $P-a+1 > P-Q$. Under the equality condition no extra shift is possible.

For $a \leq B+1$, every point of the cell is reachable from every other and $c_j = a$, proving the first identity. For $a \geq B+1$, $c_j \geq B+1$. The sum of the deficits $(2B+1)-c_j = (B-j)_+ + (B-a+1+j)_+$ is $B(B+1)$. Dividing each deficit by at least $B+1$ proves the envelope. When $a \geq 2B+1$, only the B positions at each end contribute; their denominators are $B+1,\dots,2B$. Summing both ends gives (42). □

The case of one quotient cell is not overlooked: if $Q = P$, $V_B = 0$ for every noise radius, even though the no-wrap upper bound may be positive. Exact equality in (40) is not asserted there for $B > 0$.

For a finite mixture $\chi = \sum_B \lambda_B \,\mathrm{Unif}([-B,B] \cap \mathbb{Z})$, convexity of $1/x$ gives

$$V_\chi \leq \sum_B \lambda_B V_B^\sharp. \tag{43}$$

Every finitely supported symmetric unimodal integer law has this form, with $\lambda_B = (2B+1)(\chi(B) - \chi(B+1))$; the weights telescope to the prescribed masses and sum to one. These finite-mixture assertions are the only mixture claims needed here.

Evaluating a harmonic sum up to $2B$ can require $O(B)$ additions. This is not polynomial in $\log B$. In contrast, the envelope (41), combined with the two cell widths, can be evaluated with a fixed number of integer operations on the parameters. The paper does not hide an exponential enumeration of a large cell or large noise support inside a purported efficient reduction.

13. What a recovery reduction actually obtains

Theorem 13 — Conditional transfer of true-secret recovery. In the setting of Theorem 11, let $\mathcal{A}$ be an efficient algorithm defined on every public rounded input. Suppose its success in recovering the originally drawn secret from $(\mathbf{A}, F(Z))$ is α. The procedure receiving the raw noisy observation $(\mathbf{A}, Y = [Z + E]_P)$, computing $F(Y)$, and running $\mathcal{A}$ has success at least

$$\beta \geq \frac{\alpha^2}{C_\chi}. \tag{44}$$

A larger explicitly justified upper bound on $\mathcal{R}_2$, such as that using $V_B^\sharp$, may replace C_χ.

Proof. Include X in both analytical laws and append $\mathcal{A}$'s independent random coins. The event that its output equals X has probability α under D_0 and β under D_χ. Lemma 9 and Theorem 11 give (44). The actual procedure receives only the public input; it neither knows X nor evaluates its own success event. Rounding and one oracle call give the computational overhead. □

Thus a solver for rounded recovery can be used to solve a particular noisy recovery problem with the same public and secret laws. Difficulty would transfer in that direction only if that noisy problem were already founded and C_χ had acceptable growth. A reduction between two easy problems is still a valid reduction. This paper does not supply the missing external hardness premise.

The number tL is fixed before sampling and is part of the bound. If $tL\gamma V_\chi/M \leq c \log \lambda$ for a fixed constant c, then $\log C_\chi \leq c \log \lambda$, so $C_\chi \leq \lambda^c$. This controls the analytical loss; all dimensions, encodings, and sampling costs must separately have the required polynomial bounds.

A compatible preimage is not automatically the originally drawn secret. Define compatible success to mean that the returned value $\widehat{x}$ belongs to the declared secret support and satisfies

$$F([f(A, \widehat{x})]_P) = U.$$

If, in addition, the probability that (A, U) has more than one compatible secret in that support is at most η, a solver with compatible-success probability α_{comp} has true-secret success at least $\max\{0, \alpha_{\mathrm{comp}} - \eta\}$. Equation (44) then uses that lower bound. No collision bound or efficient inverse for general vector publications is proved here.

Likewise, event preservation can be used for a publicly generated verification task, provided the reduction can generate the target and check the answer with the allowed information. If its ideal success is β_0, the resulting lower bound on an oriented decision gap is only $\alpha^2/C_\chi - \beta_0$. This may be negative when β_0 is near one half. One cannot replace it by $(\alpha - \beta_0)^2/C_\chi$ without a different proof.

A loss illustration, not security parameters. For full-period public coefficients, unit endpoint secrets, $P = 65536, Q = 256, L = 32, t = 1$, and independent uniform errors in $\{-1, 0, 1\}$, Proposition 12 gives $V_B = 256$. Hence

$$C_\chi = (257/256)^{32} < 8/7, \qquad \beta > 7\alpha^2/8 \quad \text{when } \alpha > 0. \tag{45}$$

The strict rational comparison is checked in the code. This example measures a reduction loss only; it makes no claim of 128-bit security or useful decoding parameters.

14. Decision transfer needs a separate additive estimate

Proposition 14 — Boundary coupling under the structured public law. Keep the public and secret hypotheses of Theorem 11, but allow correlations between error coordinates. Assume the entire error vector is independent of $(\mathbf{A}, X)$, with finite first absolute moments. Then

$$\mathrm{TV}\big((\mathbf{A}, F(Z)), (\mathbf{A}, F(Z+E))\big) \le \delta := \min\left\{1, \frac{\gamma J}{M}\sum_{i=1}^{tL}\mathbb{E}|E_i|\right\}. \tag{46}$$

Proof. For a fixed integer shift e, at most $\min(P, J|e|)$ canonical residues change quotient. To see this, move one unit at a time around the residue cycle; each unit step has at most J quotient boundaries. A union bound over $|e|$ steps gives the claim, including shifts exceeding a full period. For $J = 1$, there are no actual boundaries and the displayed bound is merely loose.

Conditional on the secret and previous public coefficients, a row residue has mass at most γ/M. Independence of the full error vector from the public and secret permits applying this bound at each fixed error. Averaging and summing over coordinates bounds the probability that the coupled rounded vectors differ. The common-input coupling bounds total variation, proving (46). No product of retention probabilities is used. □

Let the raw null source use a fresh uniform residue vector independently of the same public information. Define its rounded null law by actually applying F, including short final cells. Rounding the raw source and composing a target distinguisher gives

$$\mathrm{Adv}_{\mathrm{rounded}}(D) \le \mathrm{Adv}_{\mathrm{raw\ noisy}}(D \circ F) + \delta. \tag{47}$$

The null simulation is exact; only the real-world discrepancy incurs δ. In the example (45), (46) yields $\delta \le 1/12$, a constant, not a negligible asymptotic guarantee. A favorable recovery-loss constant therefore does not by itself establish decisional LWR security or IND-CPA security. A full encryption proof must still include the complete ciphertext and any reconciliation, derivation, or confirmation data.

15. Public translations must be applied at the correct stage

A public change of secret origin can be exact before rounding. For a known vector c, $X = S + c$ implies

$$f(A, X) = \mathsf{H}(A)S + \mathsf{H}(A)c$$

when $f(A, \cdot)$ is that linear map. Adding the known $\mathsf{H}(A)c$ to an unrounded sample translates the secret law exactly. It does not turn a Gaussian law into a uniform box, change the secret dimension, or preserve an unspecified conditioning event.

Proposition 15 — Exactly which scalar translations descend to the quotient. Suppose $P = QH$ with $H \ge 2$ and $Q \ge 1$. For an integer c, a deterministic map g_c satisfying

$$g_c(F(z)) = F(z + c) \quad \text{for every integer } z \tag{48}$$

exists if and only if $Q \mid c$. In that case,

$$g_c(u) = [u + c/Q]_H. \tag{49}$$

Proof. The sufficient direction is the addition identity for a shift that is a multiple of Q. For necessity write $c = kQ + r$, $0 \le r < Q$. If $r > 0$, the inputs $z_0 = 0$ and $z_1 = Q - r$ both have

quotient zero. Their translated quotients are $[k]_H$ and $[k+1]_H$, distinct because $H \geq 2$. Thus one input to g_c would require two outputs. If $Q = 1$, every integer shift meets the divisibility condition. □

When $H = 1$, every quotient is zero and every translation descends; that excluded case must not be folded into the necessity statement. For $Q \nmid P$, (5) must be used rather than (49). This elementary criterion explains a recurrent simulation error: a correct translation of a full-modulus observation is not automatically implementable after its low-order information has been discarded.

16. Exact checks and reproducibility

The accompanying standard-library Python code uses integers and `Fraction`. It implements finite laws explicitly for testing; most such enumerations are exponential in bit-encoded dimensions and are not cryptanalytic algorithms. The only randomized generation in the verifier selects reproducible rational test laws with seed `2026110808`. It does not estimate probabilities by Monte Carlo or implement a production sampler.

Verification family	Extent executed
Signed perturbations and multiple modular wraparounds	357,219
Uniform quotient/remainder profiles, including incomplete cells	1,600
Joint lifting identities with public side information	56
Translation-invariance noise profiles	350
Scalar smoothing input cases	14,756
Odd-cell minimax equalities	181
Dither comparisons retaining the complete transcript	35
Kernel and Rényi profiles	250
Event inequalities inside those profiles	4,000
Modular retention-cost profiles	5,740
Harmonic cell-cost identities and envelopes	1,260
Finite noise mixtures	80
Quotient-translation characterizations	1,260
Public-ribbon/secret states in five structured profiles	10,660

The correlated-remainder witness and the support-loss counterexamples are also checked exactly. Counts of states, events, and profiles describe different objects and must not be added into a total of independent experiments.

The five structured profiles use $d = 2$, public coefficients uniform on $[c, c+M) \cap \mathbb{Z}$, and independent uniform secret coordinates on the listed alphabet. "U1" means uniform integer noise on $\{-1, 0, 1\}$; CBD1 has weights $1, 2, 1$ out of 4. They are test profiles, not historical reconstructions or security recommendations.

(P, Q)	(c, M, L, t)	Secret alphabet / noise	Exact joint $\mathcal{R}_2$	Bound (36)
(7,3)	(4,4,2,1)	$\{1, 2\}$ / U1	625/256	4
(8,4)	(0,8,2,1)	$\{1, 3\}$ / CBD1	49/36	49/36
(11,4)	(5,6,3,1)	$\{1, 2\}$ / U1	86569/41472	27/8
(5,2)	(3,3,2,2)	$\{1, 2\}$ / U1	3281/288	2401/81
(8,3)	(6,4,2,1)	$\{0, 2\}$ / U1	507/256	49

The second profile attains the full-period equality (38). The fourth retains a shared secret across two independent ribbons. The fifth includes a nonunit or zero endpoint, giving a weak but valid bound. A finite lexicographic-preimage oracle is used only to check (44) numerically; it enumerates secrets and is not a new efficient vector recovery method. Exact recovery probabilities and public distances are stored in `A2_RESULTS.json`.

A separate script, `verify_a2_independent.py`, does not import the mathematical module. It evaluates structured products by full polynomial convolution with the secret reversed, then expands all noise vectors. Across the same five profiles it checks 399,204 noisy outcomes and reproduces every stored joint divergence, public divergence, public total variation, and recovery probability. This is a differently organized implementation check on the same states, not external peer review and not a second count of distinct cryptographic instances. It also checks 4,488 fresh-coefficient mass profiles, 19,900 boundary-cardinality profiles, and 15 exact unimodal-mixture reconstructions.

From the extracted folder run:

```
sh build_and_verify.sh
```

The two mathematical result files are deterministic and are regenerated by the public package. PDF compilation is separate from the mathematical checks. The supplied `compile_pdf.sh` compiles the delivered LaTeX source directly with `pdflatex`; Pandoc is not required to rebuild this PDF. The manifest records filenames and SHA-256 digests. No third-party paper is redistributed. No Lean formalization, lattice-attack benchmark, asymptotic parameter certification, or exhaustive novelty search is claimed.

17. What this framework contributes to future constructions

The paper supplies a vocabulary of exact identities, statistical losses, and computational obligations. It does not add a new encryption scheme and does not modify any existing construction.

Proposed step	What can be concluded	What must not be inferred
Write $QU = f(A, X) - \rho \bmod P$	Exact arithmetic relation	Independent LWE noise
Fill the quotient cell uniformly	Exact public quotient; criterion (12)	Restoration of an arbitrary original joint law
Add subtractive dither	Independent error after the stated marginalization	Independence from the published dither
Add broader noise to a cell center	Approximate channel with loss (18)	A useful hard source or the desired reduction direction
Bound joint $\mathcal{R}_2$	Recovery-success lower bound α^2/C	Negligible statistical distance or a general CPA proof
Bound joint TV in both worlds	Additive distinguisher transfer	Difficulty of the underlying source without a premise
Reverse a secret or translate it before rounding	An exact algebraic and distributional change when specified	Equality of different secret dimensions or priors

Agreement, confidentiality, recovery, and decision security remain different questions. Nothing here proves a general multidimensional inverse sampler, a necessary enumeration lower bound, or security of a vector scheme.

Any construction invoking these lemmas must first identify its precise coefficient ring or truncation operation, public transcript, secret law, error law, and conditioning variables. The present lemmas may be used only after those hypotheses are matched exactly. A future Hankel/MP-LWE hardness reduction, if available, would be a separate premise rather than a consequence of the distributional analysis here.

Scientific status. Fifteen numbered statements are accompanied by finite-space or arithmetic proofs and exact reproducibility checks. Classical antecedents are publicly cited. No claim of IND-CPA, IND-CCA, concrete security, formal machine verification, or worldwide historical priority follows from this article.

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